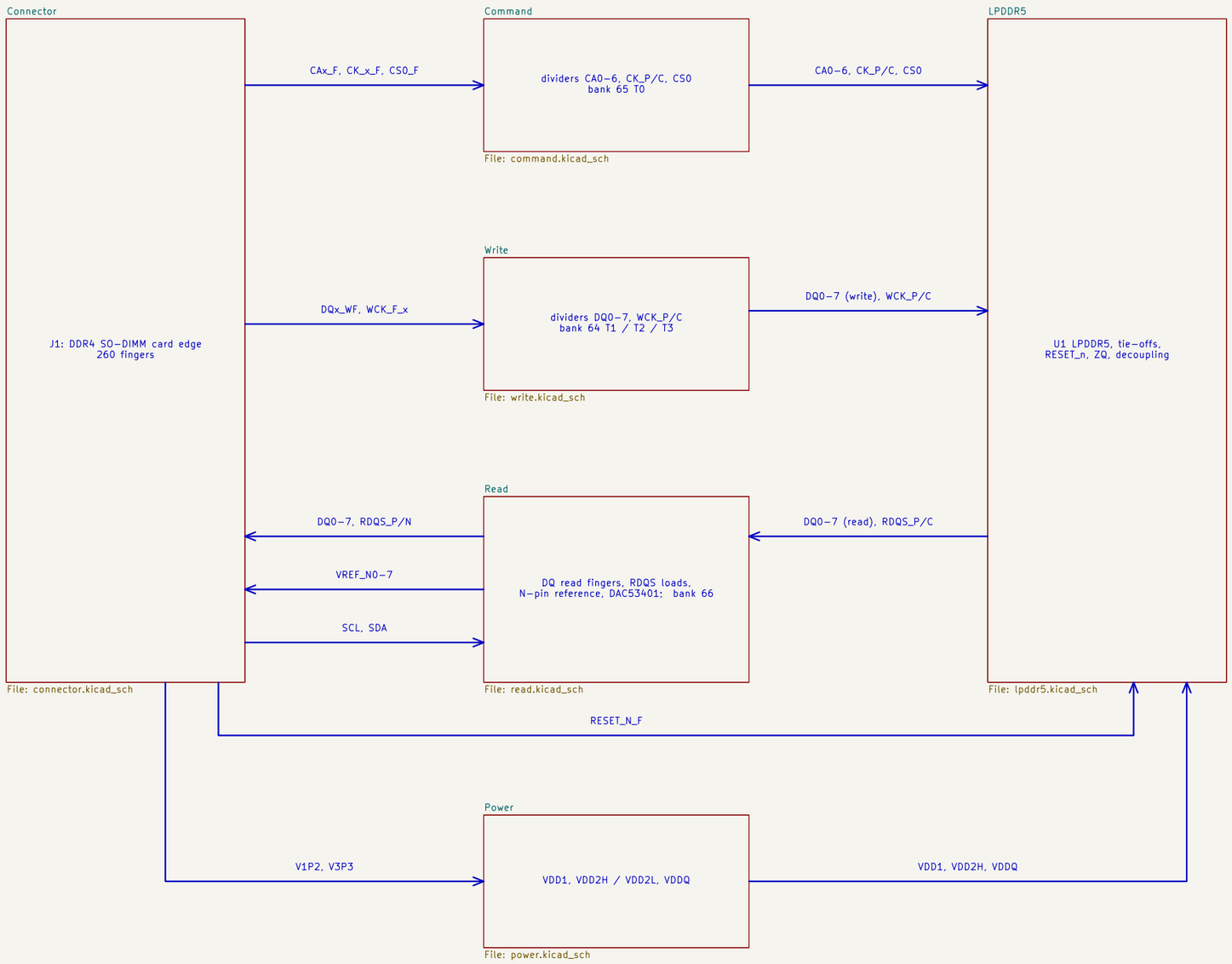
LPDDR5 testbed for ZCU104

One LPDDR5 SDRAM on a DDR4 SO-DIMM card for the ZCU104 PL memory socket: channel A, byte 0, rank 0.
Commands and writes go through resistive dividers at the card fingers; reads go straight to the FPGA, pseudo-differential against a DAC reference.



Net names

X_F, X_WF: finger side of a divider (FPGA output)
X: ball side (U1): DQx and RDQS_x also reach their
read finger (FPGA input, P pin): no series part
VREF_Nx: N finger of read bit x, 39 R into VREF_RD
V1P2, V3P3: host rails; VDD1, VDD2H, VDDQ: card rails

All nets are global labels or power symbols; the sheets
have no hierarchical pins.

Design data

design/netlist.json: parts, pins, fingers, delay targets,
stackup, rules, floorplan (single source of truth)
design/DESIGN.md: rationale
scripts/gen_schematic.py: writes these sheets from the netlist
scripts/check_schematic.py: ERC and pin-by-pin netlist parity

131 parts, 119 in the BOM, none DNP.
Host: ZCU104 rev 1.0, XCZU7EV-2FFVC1156,
socket J1 (HP banks 64 / 65 / 66 at 1.2 V)

Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: overview

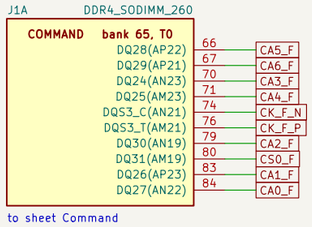
Sheet: /
File: lpddr5-testbed.kicad_sch

Title: LPDDR5 testbed for ZCU104

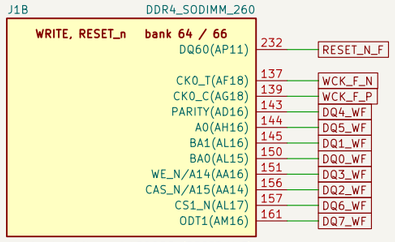
Size: A3	Date: 2026-11-10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/7

J1: DDR4 SO-DIMM card edge, 260 fingers

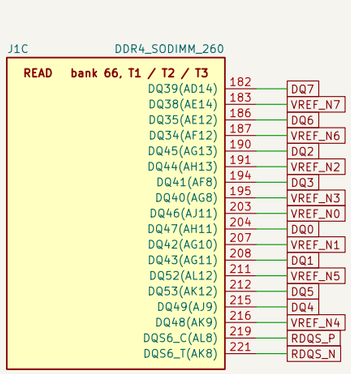
Plugs into the ZCU104 PL DDR4 socket. Pin names: DDR4 SO-DIMM signal name (FPGA ball of the XCZU7EV). Odd fingers on the front, even fingers on the back.



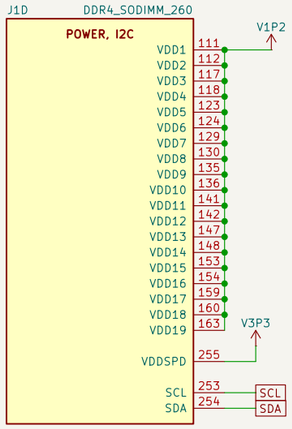
to sheet Command



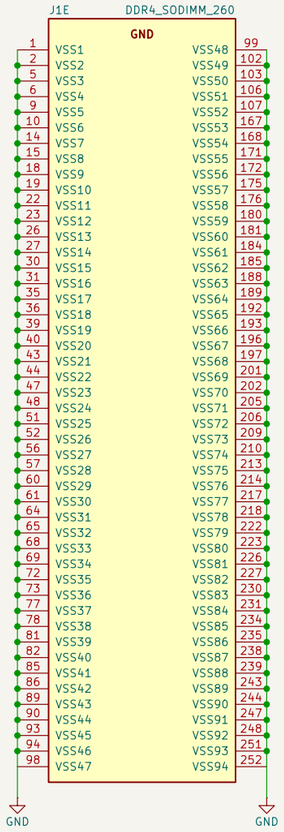
to sheets Write and LPDDR5 (RESET_N_F)

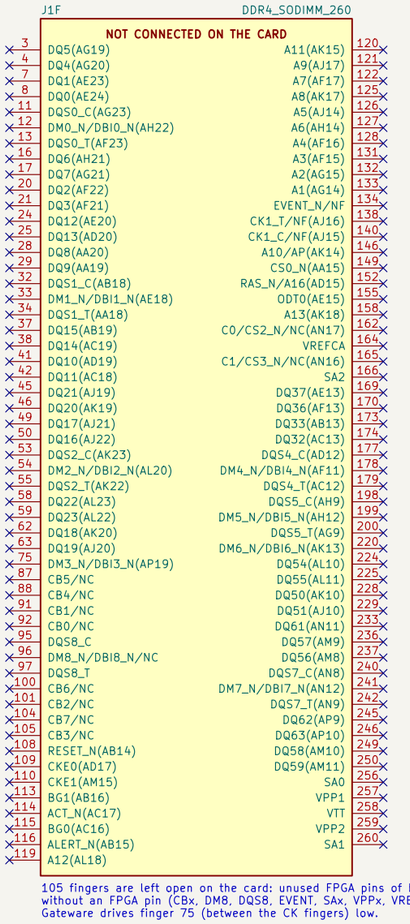


read fingers: DQx (the net of the divider node, sheet Write) = P pin,
VREF_Nx = N pin of the same differential input (sheet Read); RDQS_P / RDQS_N = true pair



19 VDD fingers: host 1.2 V
VDDSPD: host 3.3 V
SCL / SDA: host I2C (4.99 k pull-ups to 3.3 V on the ZCU104), reference DAC at 0x48





105 fingers are left open on the card: unused FPGA pins of banks 64 / 65 / 66, and the fingers without an FPGA pin (CBx, DM8, DQS8, EVENT, SAx, VPPx, VREFCA, VTT).
Gateware drives finger 75 (between the CK fingers) low.

Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: Connector

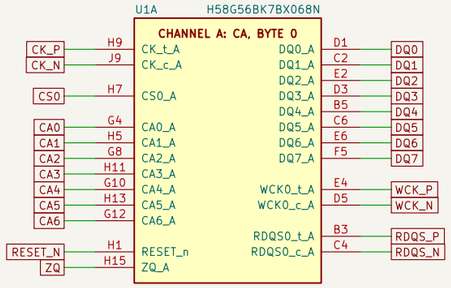
Sheet: /Connector/
File: connector.kicad_sch

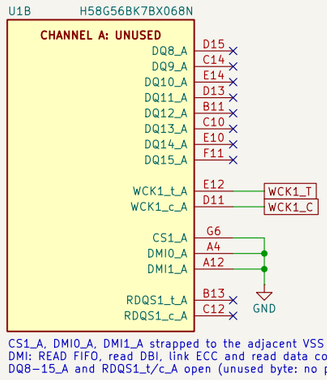
Title: LPDDR5 testbed for ZCU104

Size: A3	Date: 2026-11-10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 2/7

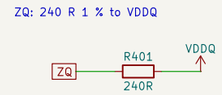
U1: LPDDR5 SDRAM, channel A / byte 0 / rank 0 in use

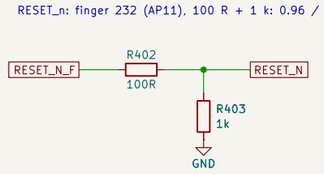
SK hynix H58G56BK7BX068N (JEDEC 315-ball 2CHx32 ball map). Channel B and the upper byte of channel A are tied off or left open as listed below.

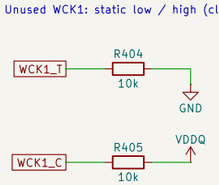


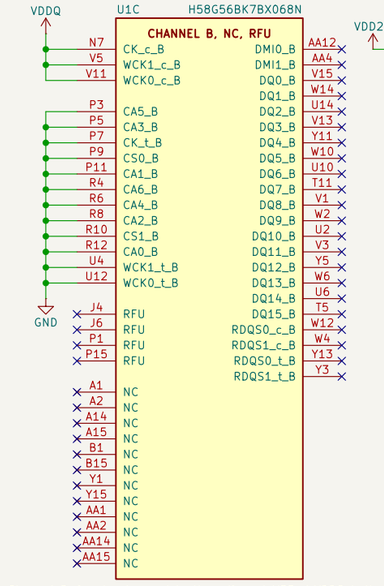


CS1_A, DMI0_A, DMI1_A strapped to the adjacent VSS ball on F.Cu.
DMI: READ FIFO, read DBI, link ECC and read data copy must stay off.
DQ8–15_A and RDQS1_t/c_A open (unused byte: no pull-downs, DESIGN.md section 5).

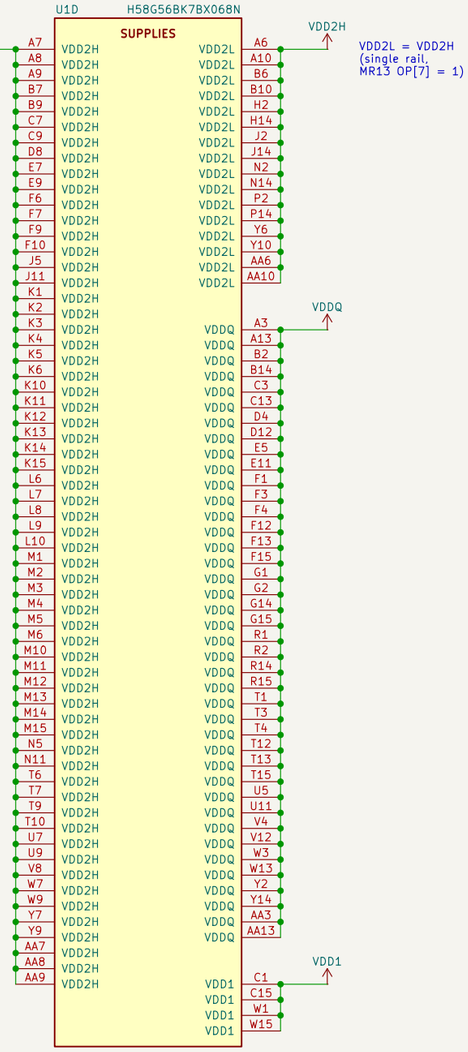


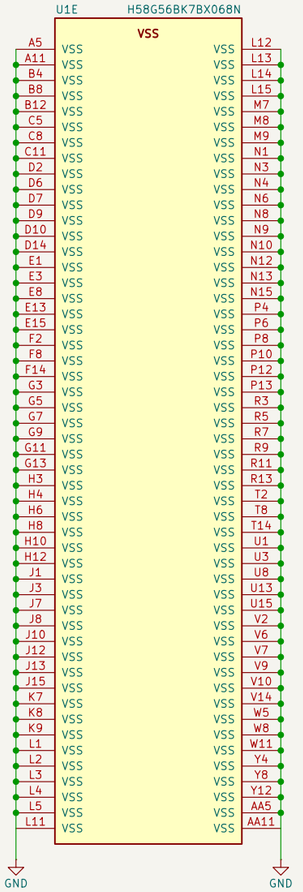




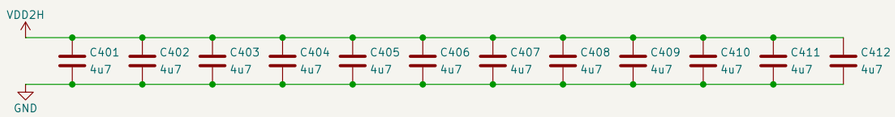


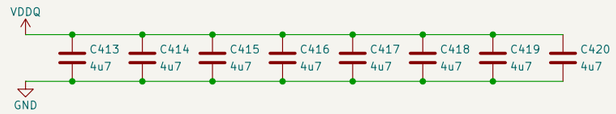
Channel B: inputs strapped to the adjacent VSS / VDDQ ball (CK_c_B through its via-in-pad), DQ / DMI / RDQS open.

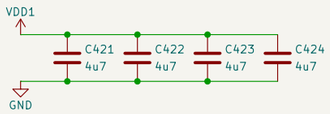




Decoupling on B.Cu under U1, on the via-in-pad supply balls







Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: LPDDR5

Sheet: /LPDDR5/
File: lpddr5.kicad_sch

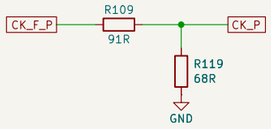
Title: LPDDR5 testbed for ZCU104

Size: A3	Date: 2026-11-10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 3/7

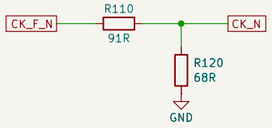
Command bus: resistive dividers at the card fingers

ZCU104 bank 65 byte group T0 (1.2 V SSTL12 outputs) -> LPDDR5 channel A CA / CK / CS at VDDQ level. Series resistor at the finger, shunt at the In3 node via.

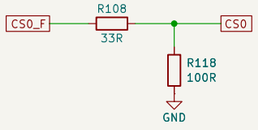
CK_P: finger 76 (AM21) -> U1.H9



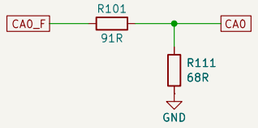
CK_N: finger 74 (AN21) -> U1.J9



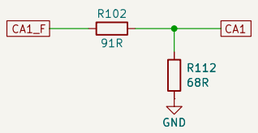
CS0: finger 80 (AM19) -> U1.H7



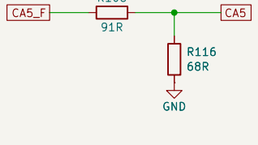
CA0: finger 84 (AN22) -> U1.G4



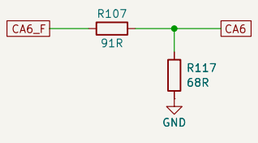
CA1: finger 83 (AP23) -> U1.H5



CA2: finger 79 (AN19) -> U1.G8



CA3: finger 70 (AN23) -> U1.H11



CA4: finger 71 (AM23) -> U1.G10

CA5: finger 66 (AP22) -> U1.H13

CA6: finger 67 (AP21) -> U1.G12

Levels at the ball (min / nominal / max)

Corners: FPGA driver 26 / 36 / 44 ohm, VCCO 1.14 / 1.20 / 1.26 V, resistors 1 %.

CA0-6, CK_P, CK_N: 91 R / 68 R 0.378 / 0.418 / 0.468 V pin current 6.9 mA max
back impedance 44.6 ohm into the 45.6 ohm In3 line; limit: DC level <= 0.47 V target, <= 0.488 V
(card VDDQ min) hard; >= 0.351 V at boot (default VREF(CA) 0.256 V max + 0.095 V)

CS0: 33 R / 100 R 0.640 / 0.710 / 0.797 V pin current 8.0 mA max
limit: 0.55 V (ViHPD) .. VDD2H + 0.2 V; Vref_CS = VDD2H / 3 = 0.35 V

FPGA: SSTL12 / DIFF_SSTL12, RDRV_40_40, FAST.
CK_P on finger 76 (P pin), CK_N on finger 74 (N pin); finger 75 between them stays open.
All 10 nets are delay matched FPGA die pad to ball within +/-2 ps (CK_P to CK_N +/-1 ps)
on In3 between two ground planes; see design/DESIGN.md section 1.

Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: Command

Sheet: /Command/
File: command.kicad_sch

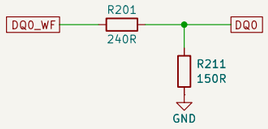
Title: LPDDR5 testbed for ZCU104

Size: A3	Date: 2026-11-10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/7

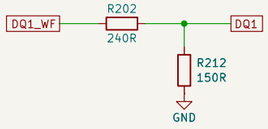
Write path: resistive dividers at the card fingers

ZCU104 bank 64 byte group T1 / T2 / T3 (1.2 V outputs) -> LPDDR5 channel A byte 0 DQ and WCK0. The DQ shunt is also the read load; the read branch leaves the same node without a part (sheet Read).

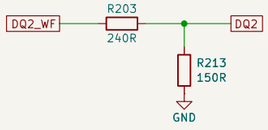
DQ0: finger 150 (AL15) -> U1.D1



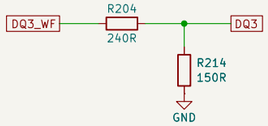
DQ1: finger 145 (AL16) -> U1.C2



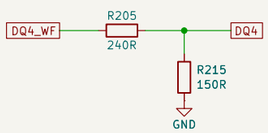
DQ2: finger 156 (AA14) -> U1.E2



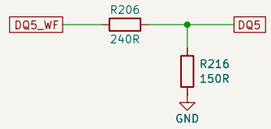
DQ3: finger 151 (AA16) -> U1.D3



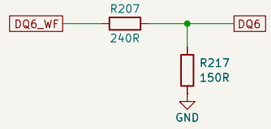
DQ4: finger 143 (AD16) -> U1.B5



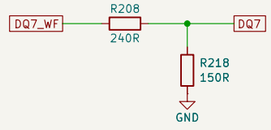
DQ5: finger 144 (AH16) -> U1.C6



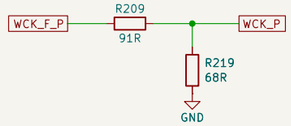
DQ6: finger 157 (AL17) -> U1.E6



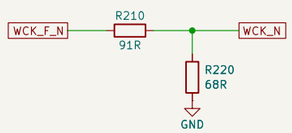
DQ7: finger 161 (AM16) -> U1.F5



WCK_P: finger 139 (AG18) -> U1.E4



WCK_N: finger 137 (AF18) -> U1.D5



Levels at the ball (min / nominal / max)

DQ0–7: 240 R / 150 R   0.389 / 0.423 / 0.460 V   pin current 3.1 mA max
    limit: <= 0.47 V (VDDQ min); VREF(DQ) programmed to half the level (<= 0.225 V)
    node source resistance 97 ohm, tau 2.6 ns: design target 100 Mb/s writes

WCK_P, WCK_N: 91 R / 68 R   0.378 / 0.418 / 0.468 V per leg
    limit: differential >= 0.30 V p–p, also with the 40 ohm WCK ODT forced during CBT

WCK_P on finger 139 (N pin), WCK_N on finger 137 (P pin):
    the host pair is driven INVERTED by the gateware.
The write pins are Hi–Z during reads.
DQ0–7 are NOT delay matched on the card: die pad to ball +/–26.7 ps (F.Cu comb, the T at the divider); the FPGA closes it with a per–bit output delay (design/gateware_deskew.json).
WCK_P to WCK_N within +/–1 ps (B.Cu pair); see design/DESIGN.md section 14.

Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: Write

Sheet: /Write/
File: write.kicad_sch

Title: LPDDR5 testbed for ZCU104

Size: A3	Date: 2026–11–10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 5/7

Read path: DQ nodes straight to the P fingers, reference on the N fingers

LPDDR5 DQ0–7 –> divider node (sheet Write) –> ZCU104 bank 66 P pins, no series part; every N pin sits on VREF_RD (pseudo–differential). RDQS is a true differential pair.

DQ read fingers (no series part: same net as the divider node)

DQ0 –> finger 204 (AH11), P (U1.D1)

DQ1 –> finger 208 (AG11), P (U1.C2)

DQ2 –> finger 190 (AG13), P (U1.E2)

DQ3 –> finger 194 (AF8), P (U1.D3)

DQ4 –> finger 215 (AJ9), P (U1.B5)

DQ5 –> finger 212 (AK12), P (U1.C6)

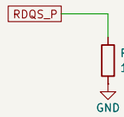
DQ6 –> finger 186 (AE12), P (U1.E6)

DQ7 –> finger 182 (AD14), P (U1.F5)

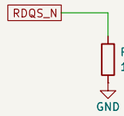
Without a series resistor the open FPGA end rings for 1 .. 2 ns after an edge:
0.54 V / −0.12 V at worst (strong driver), 0.45 V / −0.06 V nominal; the pin allows
−0.20 .. 1.40 V. Damping, if wanted: MR3 drive strength 48 / 60 ohm (gateware).

RDQS: 150 R load per leg, no series part; host pair DQS6 (fingers 219 / 221) inverted (RDQS_P on its N pin)

finger 219 (AL8) <– U1.B3

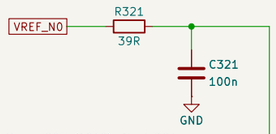


finger 221 (AK8) <– U1.C4

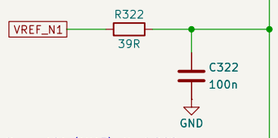


N fingers: 39 R into VREF_RD, 100 nF to GND at each resistor

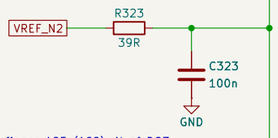
finger 203 (AJ11), N of DQ0



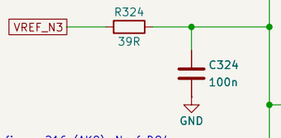
finger 207 (AG10), N of DQ1



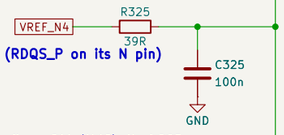
finger 191 (AH13), N of DQ2



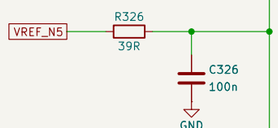
finger 195 (AG8), N of DQ3



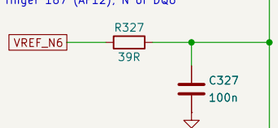
finger 216 (AK9), N of DQ4



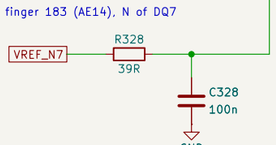
finger 211 (AL12), N of DQ5



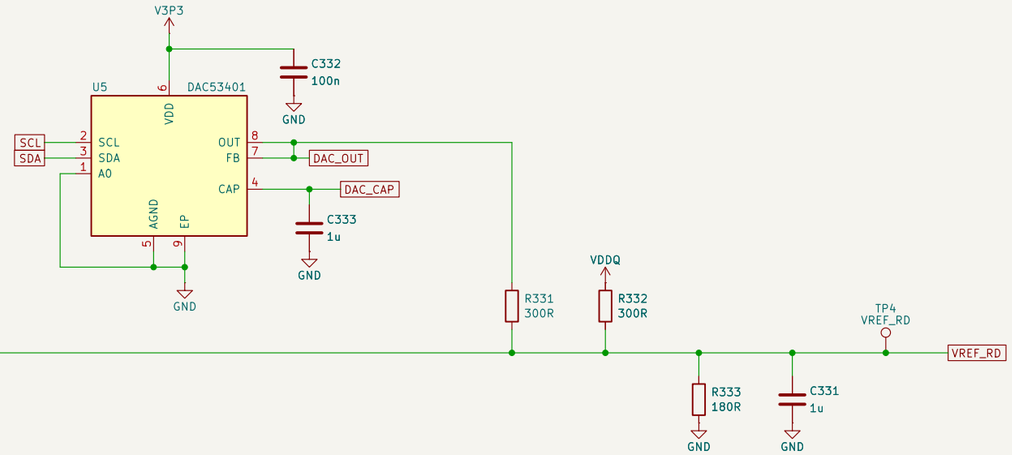
finger 187 (AF12), N of DQ6



finger 183 (AE14), N of DQ7



Reference: DAC53401 (I2C 0x48) and the VDDQ–derived power–up default



VREF_RD network: DAC_OUT –300R– VREF_RD; VDDQ –300R– VREF_RD; VREF_RD –180R– GND: 1 uF + 8 x 100 nF on the bus
DAC output Hi–Z (factory default): 0.187 V, 112 ohm source: reads work before any I2C traffic
DAC active: V = 0.2727 x Vdac + 0.1363 V (0.136 .. 0.631 V, 0.48 mV / LSB, code 121 = 0.195 V)
keep VREF_RD at or below 0.20 V if the measured read high level is above 0.42 V (VICM <= 330 mV)

Read levels (150 R shunt on sheet Write is the only DC load; no series resistor):
read high 0.374 / 0.386 / 0.405 V (JEDEC bound 0.362 .. 0.431 V), reference 0.195 V
receiver: IBUFDS_DPHY, IOSTANDARD MIPI_DPHY_DCI, DIFF_TERM = FALSE; read pins are inputs only
each N line ends in 39 R = the host line impedance, AC–grounded by 100 nF

RDQS_P / RDQS_N: VID 0.386 V, VICM 0.193 V while the DRAM drives; both legs 0 V otherwise:
gate the receiver output with the read–enable window.
DQ0–7 read paths are delay matched ball to FPGA die pad within +/–1 ps (In3 bundle).

Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: Read

Sheet: /Read/
File: read.kicad_sch

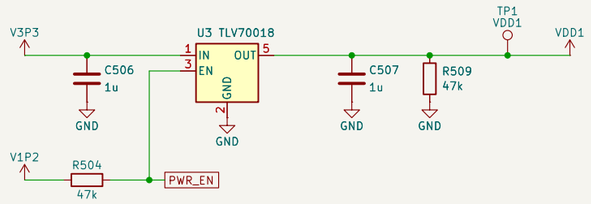
Title: LPDDR5 testbed for ZCU104

Size: A3	Date: 2026–11–10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 6/7

Power: three LDOs from the host 1.2 V and 3.3 V fingers

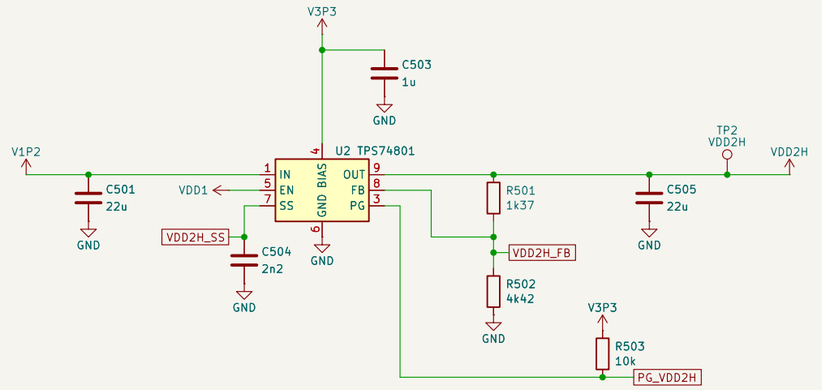
Sequence: VDD1 (enabled by V1P2) -> VDD2H = VDD2L (enabled by VDD1, soft start) -> VDDQ (enabled by the VDD2H power good).

VDD1 1.80 V (U3 TLV70018DDCR)



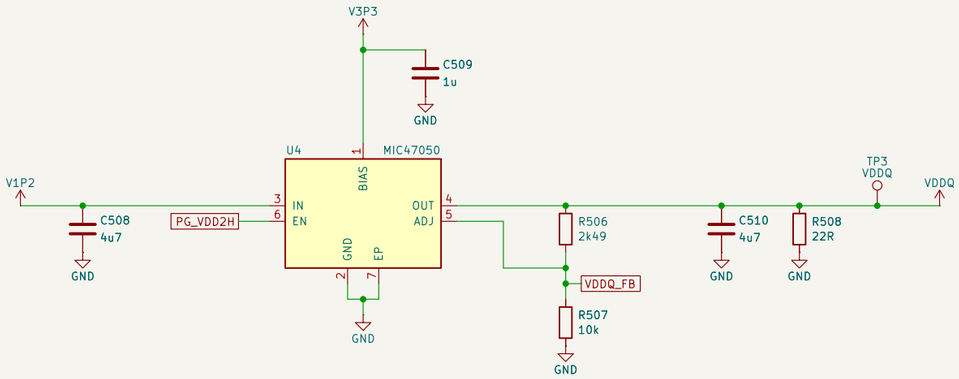
EN = V1P2 through 47 k: VDD1 cannot rise before the host 1.2 V rail.
R509 (47 k) empties VDD1 after power-off (no output discharge in U3).
U3 pin 4 (NC) is not drawn and stays open.

VDD2H = VDD2L 1.048 V (U2 TPS74801DRCRM3)



VOUT = 0.8 V x (1 + R501 / R502) = 1.048 V, +/-1 % (new die, guaranteed by the M3 suffix).
EN = VDD1. C504 (2.2 nF): 3.3 ms soft start. PG (open drain, 10 k to V3P3) enables U4.
U2 pins stacked in the symbol: 2 on 1 (IN), 10 on 9 (OUT), 11 = exposed pad on 6 (GND).

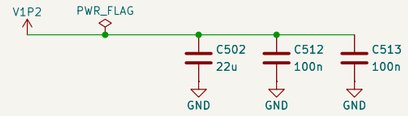
VDDQ 0.500 V (U4 MIC47050YML-TR)



VOUT = 0.4 V x (1 + R506 / R507) = 0.4996 V.
EN = PG_VDD2H: VDDQ starts when U2 reports power good.
R508 (22 R) discharges VDDQ after U4 is disabled, time constant 0.93 ms.

Host rails at the fingers

1.2 V: 19 VDD fingers, 0.77 A max (41 mA per contact)



3.3 V: finger 255 (VDDSPD), 30 mA max



GND: 94 VSS fingers



Assembly fiducials (three per face)
Power-up (JESD209-5 Table 19 / 20): nothing starts until both host rails are up;
VDD1 first, VDD2H ramps 3.3 ms behind it, VDDQ follows the VDD2H power good.
Power-off: 1.2 V first: VDD2H follows; PG disables U4; R508 pulls VDDQ down, VDD1 falls last.
Not connected: VPP (2.5 V), VTT, VREFCA. No SPD EEPROM.



Generated from design/netlist.json by scripts/gen_schematic.py
Sheet: Power

Sheet: /Power/
File: power.kicad_sch

Title: LPDDR5 testbed for ZCU104

Size: A3	Date: 2026-11-10	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 7/7